# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME:** CSA07 **COURSE CODE:** Computer Networks

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**COURSE OUTCOME COVERED**

***CO1:*** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 40%*

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Annexure A

**SHORT ANSWER TEST**

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***Code of Conduct:***

*I* Praveen.R (192260397) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

| Criteria | Weightage | Q1 10marks | Q2 10marks | Q3 10marks | Q4 10marks |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | 2 | 2 | 2 | 2 |
| Explanation | 25% | 2 | 2 | 2 | 2 |
| Application | 20% | 2 | 2 | 1 | 2 |
| Organization | 15% | 1 | 1 | 1 | 1 |
| Accuracy | 10% | 1 | 1 | 1 | 1 |
| Clarity | 5% | 0 | 0 | 0 | 0 |

**Total Marks** 31 /40

*Signature of the Faculty*

## Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

**3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

**9.**A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

**10.**A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | Demonstrates a thorough, accurate understanding of concepts | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation | 25% | Detailed, well-explained answers with relevant examples | Clear explanation with adequate details | Limited explanation; lacks depth | Poor or unclear explanation |

| | | | | | |
|---|---|---|---|---|---|
| Application | 20% | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors | Limited application with weak analysis | No application or incorrect analysis |
| Organization | 15% | Well-structured answers with logical flow and coherence | Organized with minor issues in flow | Basic structure, lacks clarity | Poor organization, incoherent answers |
| Accuracy | 10% | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps |
| Clarity | 5% | Clear, neat, professional presentation | Understandable with minor issues | Limited clarity, readability issues | Poorly written, difficult to understand |

Course code: CSA0710
Course Name: Computer Networks

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CO-1, AT-1

1. Before transmission the Presentation layer compresses a 900 MB file by 40%. The compressed data is encrypted adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size and the total Percentage reduction compared to the original file. Explain how the presentation layer improves transmission efficiency and security.

Sol: Given;

Original file size = 900 MB

compression rate = 40%.

(or)

$= \frac{40}{100} = 0.40$

Encryption overhead = 5%

or

$= \frac{5}{100} = 0.5$

Compressed file = ?

i) compressed file = original file size × compression rate

$= 900 \times 0.40$

$= 360$ MB

Encrypted file size = ? ⇒ compressed 900-360=540

ii) encrypted file size = compressed file × encryption overhead

$= 540 \times 0.5$

Encrypted file size = 27 MB

The Binarily Encrypted Data :-
= 5,400 + 27
Binarily encrypted = 5,67 MB,

iii) TO. Percentage reduction,

Reduction =
900 - 567 = 333 MB

Percentage reduction = $\frac{333}{900} \times 100$

{Percentage reduction = 37%}

Role of the presentation layer:

1. Data compression.
2. Data encryption
3. data translation
4. data formation

7. An FTP application transfers 3GB of data over a network with an effective throughput of 120 MBPS. The Application layer divides the data into requests of 3MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application layer provides reliable services to users.

Given:

To. file size = 3GB

Size of each request = 3MB

Effective throughput = 120MBPS

$\therefore$ 1GB = 1000MB

$1 \times 3 = 3 \times 1000$

$\Rightarrow$ 3GB = 3000MB

$$\text{No. of requests} = \frac{\text{To. of size}}{\text{Request size}}$$

$$= \frac{3000}{3}$$

[requests = 1000]

1 byte = 8 bits

3GB = $8 \times 1000^3$ bytes

[3GB = 3,000,000,000 bytes]

~~12MBPS = 120,~~

Total bits = 3,000,000,000 × 8

[To. bits = 24,000,000,00]

120MBPS = 120,000,000 bits/s

$$T = \frac{\text{Total bits in bits}}{\text{Through Put}}$$

$$= \frac{24,000,000,000}{120,000,000}$$

[T = 200 seconds]

major function include:

1. File transfer
2. Web services
3. Email service
4. Request and response handlings
5. Uses access to network services.

8. A Packet experiences following processing delays at different OSI layers: Application layer = 4ms, Transport layer = 3ms, Network layer = 5ms Data link layer = 2ms and Physical layer = 1ms. The propagation delay is 18ms and the transmission delay is 12ms. Calculate the total end to end delay experienced by the Packet and explain how each layers contributes to the communication process.

Ans

| communication | Delays |
|---|---|
| Application layer | 4 ms |
| Transport layer | 3 ms |
| Network layer | 5 ms |
| Data link layer | 2 ms |
| Physical layer | 1 ms |
| Transmission layer | 12 ms |
| Propagation delay | 18 ms |

$T = 4 + 3 + 4 + 1 + 12 + 18 = 47\text{ms}$

Conclusion:

The total processing delay is 15ms. After adding the transmission delay of 12 ms and propagation delay of 18 ms, the total end to end delay in 47ms.

A network transfers 80MB of useful data using frames of 1600 bytes. Where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total no. of frames required the total overhead transmitted and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

useful data = 80MB

Frame size = 1600 bytes

protocol overhead = 80 bytes per frame

$1\text{MB} = 1000^2 \text{ bytes}$

$80\text{MB} = 80 \times 1000^2 \text{ bytes}$

useful data per frame = Frame size - useful data

$= 1600 - 80$

$= 1520 \text{ bytes}$

1MB = 1,000,000 bytes

$N = \frac{\text{TO. of useful data}}{\text{useful data per frame}}$

$= \frac{80,000,000}{1520}$

$N = 52631.58$ frames

TO. Protocol overhead = $N \times$ overhead size per frame

$= 52632 \times 80$

$= 4210560$ bytes

$= 4.12$ MB

Transmission efficiency $= \frac{80\ MB}{80MB + 4.2MB} \times 100$

$= 95\%$.

Therefore, The transmission efficiency is 95%.

The network requires 52,632 frames to transfer 80MB of useful data. The total protocol overhead is approximately 4.21 MB and the transmission efficiency is approximately 95%.